

Exercise Sheet

Task 1 – Flipping Bits in Assembly

Write an assembly program that flips the LSB of the value saved in register t0. Use the documentation provided in the Moodle course.

What do you have to change to flip the second last bit?

```
xori t0,t0,1
```

To flip bit n we use the immediate $2^{(n-1)}$ instead of 1.

Task 2 – Two's Complement in Assembly

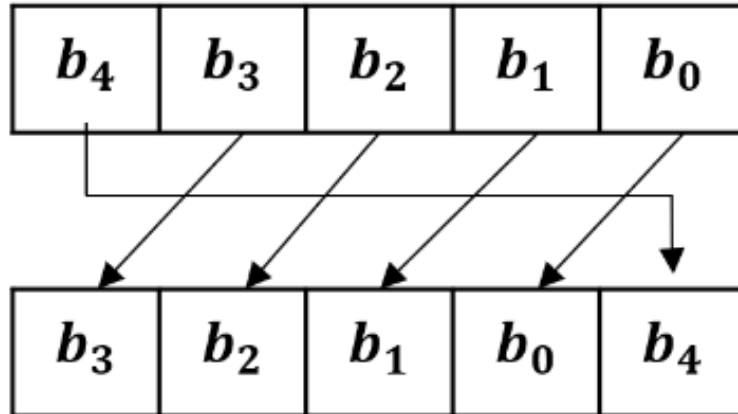
Write a program to compute the two's complement representation of a number. You can assume, that the amount of said number is already in register t0. You do not have to implement an overflow check at this time. Use the documentation provided in the Moodle course.

```
xori t0, t0, -1
```

```
addi t0, t0, 1
```

Task 3 – Rotations

You already learned what shifts are in the lecture. Here we want to perform a rotation, a shift where the bit is pushed out at one end and then inserted back into the register at the other. There are no RISC V assembly instructions to perform rotations, therefore it will be your task to write a short program that implements the functionality of a logical rotation to the left, as can be seen here:



```
SRLI t1, t0, 32-n  
SLLI t0,t0,n  
OR t0,t0,t1
```

First, we put the n highest bits in a temporary register so that they don't get deleted when we perform a shift by n bits to the left in the second instruction. We use a right shift for that, so that the n highest bits are already in the right position in the temporary register. Now we have our register from the beginning with the $32-n$ highest bits in the right position and the n lowest bits set to 0 and the temporary register with the n lowest bits in the right position and the $32-n$ highest bits set to 0. We combine both in a register using OR.

Task 4 – Instruction Formats

How big is the address space for jump and branch instructions? Which address space is bigger?

Jump-Instructions use the UJ instruction format and therefore have 20 bits for the immediate offset.
Branch-Instructions use the SB instruction format and therefore have 12 bits for the immediate offset.
Thus, the Jump-Instructions have a bigger address space.